

Descriptive Statistics and Introduction to Probability
Problem sheet N°1

Exercise 1. Among these assertions, specify which ones are true and which ones are false.

1. We call variable, a characteristic that we study.
2. The task of descriptive statistics is to collect data.
3. The task of descriptive statistics is to present data in the form of tables, graphs and statistical indicators.
4. In Statistics, variables are classified into different types.
5. The values of variables are also called modalities.
6. For a qualitative variable, each statistical individual can have only one modality.
7. To perform statistical treatments, a quantitative variable is sometimes transformed into a qualitative variable.
8. In practice, when a discrete quantitative variable has a large number of distinct values, it is treated as continuous.

Exercise 2. Specify for each of these variables whether it is ordinal or nominal qualitative data, discrete or continuous quantitative variable.

Area of an apartment - Number of subjects acquired in a semester - Eye color - Grades in higher education - Cholesterol level - Speed - Height - Type of car - Number of defective parts made by a machine - Number of languages spoken - Annual income - Citizenship - Distance - Place of residence - Age.

Exercise 3. For the following topics of study specify: the population, the statistical unit, the statistical variable and its nature.

1. Study of the life span of electric lamps of a brand ALPHA.
2. Study of the absenteeism of the workers, in days, of a factory.
3. Distribution of students according to the mention obtained at the Baccalaureate.
4. Number of accidents involving two cars on a set of 100 road intersections chosen at random in a city.

Exercise 4. The distribution of 1600 students in a high school according to the foreign language studied is given in the following table :

Languages	German	English	Spanish	Italian	Various
Size	351	934	205	69	41

1. Give the population and the character studied by specifying its nature.
2. Represent graphically the distribution of the 1600 students according to the language studied by the appropriate diagrams.

Exercise 5. 1. Give one example of a situation in which
 (i) the mean is an appropriate measure of central tendency.
 (ii) the mean is not an appropriate measure of central tendency but the medians is an appropriate measure of central tendency.
 2. The rate of inflation in fire successive year in a country was 5%, 8%, 12%, 25% and 34%. What was the average rate of inflation per year?

Exercise 6. The temperature (in °C) recorded each day in a given city was measured during the entire month of January and the following results were obtained:

12,5 – 14,2 – 15,6 – 14,5 – 13,3 – 11,9 – 10,7 – 10,4 – 9,9 – 9,4 – 10,8 – 14,9 – 15,6 – 15,8 – 14,3 – 13,6 – 12,9 – 13,6 – 16,2 – 16,8 – 17,6 – 17,0 – 15,3 – 14,2 – 12,6 – 11,8 – 11,4 – 10,2 – 10,0 – 9,1 – 8,8.

1. Give the population and the character studied by specifying its nature.
2. Determine the range of this series. Establish the statistical table of this distribution.
3. Complete the statistical table with the frequencies, the percentages and the increasing and decreasing cumulative numbers.
4. Draw the histogram as well as the curves of the increasing and decreasing cumulative numbers.
5. Determine the modal class or classes, the mean and the median.

Exercise 7. For all the soccer matches counting for the English Cup, we will study the number of goals scored by the winning team including the draws.

No. of goals scored	0	1	2	3	4	5	6	Total
No. of matches	80	145	124	75	40	10	6	480

1. What is the character studied and specify its nature.
2. Complete the statistical table with percentages.
3. Represent graphically the less than Ogive Graph and the more than Ogive Graph then locate the median.
4. Calculate the mean and median.

Exercise 8. A household survey of summer vacation budgets yielded the following results:

Budget X(10^2 DA)	Cumulative frequency
[800 ; 1000[	0.08
[1000 ; 1400[	0.18
[1400 ; 1600[	0.34
[1600 ; β [	0.64
[β ; 2400[	0.73
[2400 ; α [	1

1. Determine the value of α knowing that the range of the series is equal to 3200×10^2 DA.
2. Complete the table with the frequencies.
3. Determine the value of β in the following two cases:
 - a. The average budget is equal to 1995×10^2 DA.
 - b. The median budget is equal to 1920×10^2 DA.

Consider throughout the following β determined from the average budget.

4. Draw the histogram.
5. Calculate the median and the mode.